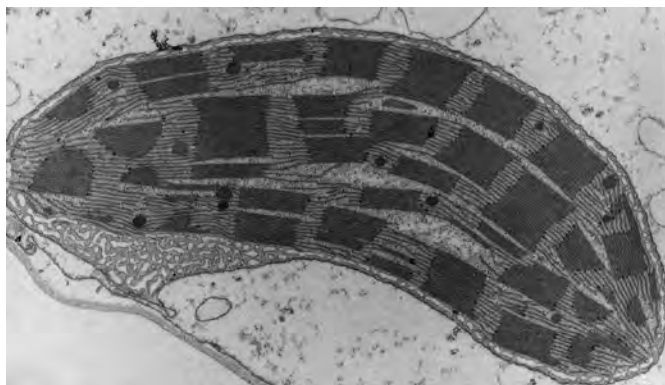


- 5 (a) The photograph shows an organelle found in many plant cells, as seen using a microscope.



(Source: © Ashley Cooper pics / Alamy Stock Photo)

- (i) Give the name and function of this organelle.

(1)

Name

Function

.....

.....

- (ii) Name the type of microscope used for this photograph.

Give **one** reason for your answer.

(2)

Name

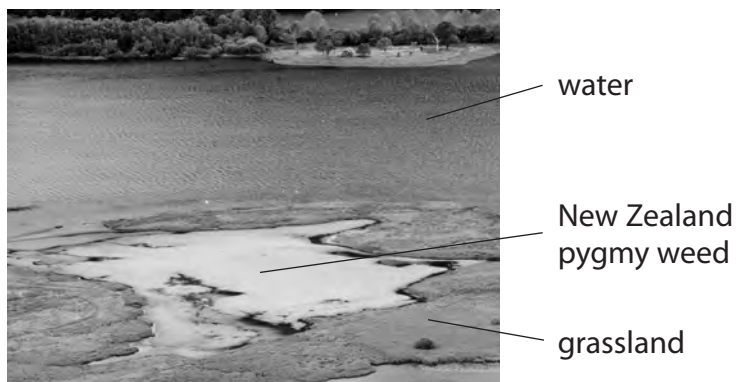
Reason

.....

.....



- (b) The photograph shows a lake partly covered with New Zealand pygmy weed plants (*C. helmsii*).



(Source: © Science History Images / Alamy Stock Photo)

The effect of New Zealand pygmy weed on the biodiversity of a lake was investigated.

Species richness and an index of diversity were determined.

- (i) State what is meant by **species richness**.

(1)

- (ii) The number of individuals of different species in this area of the lake was measured **before** the appearance of the New Zealand pygmy weed.

The table shows these results.

Species	Number of individuals (n)	(n – 1)	n(n – 1)
A	54	53	2862
B	12		
C	3	2	6
D	31	30	930
E	7	6	42
F	46	45	
	Total (N) =		$\sum n(n - 1) =$

Calculate the index of diversity (D) for this area of the lake using the formula:

$$D = \frac{N(N - 1)}{\sum n(n - 1)}$$

Use the table to help you.

Give your answer to **two** decimal places.

(3)

Answer



- (iii) A year **after** the appearance of the New Zealand pygmy weed in this lake the index of diversity (D) was found to be 1.8.

Suggest the effect of the New Zealand pygmy weed on the diversity of this part of the lake.

(2)

(Total for Question 5 = 9 marks)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

